Progressive Encephalopathy With Amyotrophy And Optic Atrophy (PEAMO): A Comprehensive Disease Characterization

Disease: Progressive Encephalopathy With Amyotrophy And Optic Atrophy (PEAMO) **MONDO:** MONDO:0014968 · **OMIM:** #617207 · **Causal gene:** *TBCE* (tubulin-specific chaperone E / tubulin folding cofactor E), HGNC:11582, chromosome **1q42.3 Inheritance:** Autosomal recessive · **Category:** Mendelian (rare neurodegenerative tubulinopathy)

Summary

Progressive Encephalopathy With Amyotrophy And Optic Atrophy (PEAMO) is caused by biallelic loss-of-function variants in *TBCE*, the gene encoding tubulin-specific chaperone E (cofactor E). TBCE is one of five tubulin-folding cofactors (TBCA–TBCE) essential for assembling the α/β-tubulin heterodimer, the building block of microtubules. PEAMO was delineated as a distinct clinical entity by Sferra and colleagues in 2016, who identified recurrent biallelic *TBCE* mutations—most notably the founder-like missense change c.464T>A (p.Ile155Asn)—in six affected children from four unrelated families originating from the same area of Southern Italy (PMID: 27666369). The disorder is a childhood-onset, progressive neurodegenerative tubulinopathy defined by a clinical triad: **progressive encephalopathy/ spastic-ataxic tetraparesis, distal spinal muscular atrophy (amyotrophy) from a distal motor axonal neuropathy, and optic atrophy**, with brain MRI showing progressive thinning of the corpus callosum and cerebro-cerebellar atrophy (PMID: 41833177).

Mechanistically, destabilized/degraded TBCE protein impairs microtubule polymerization and disrupts an ARF1/TBCE-dependent Golgi homeostasis program, producing a **distal "dying-back" axonopathy** that begins at the motor endplates and ascends to axonal loss and neuronal apoptosis. This model is strongly supported by the *progressive motor neuronopathy* (pmn) mouse, which carries a *Tbce* missense mutation (p.Trp524Gly), faithfully recapitulates the

motor neuronopathy, and is fully rescued by wild-type *Tbce* cDNA—establishing both causality and proof-of-concept for gene replacement (PMID: 12389029; PMID: 12446740; PMID: 24951541; PMID: 27488538).

PEAMO sits within a broader **allelic *TBCE* disease spectrum**. In a large aggregation of 335 *TBCE* patients, Sanjad-Sakati syndrome (hypoparathyroidism-retardation-dysmorphism, HRD) was by far the most common presentation (85%), whereas the PEAMO/neurodegenerative phenotype accounted for only ~5% of cases (PMID: 41833177). The critical distinction is that **PEAMO is a progressive neurodegenerative tubulinopathy that lacks the congenital hypoparathyroidism, intrauterine growth retardation, and facial dysmorphism** that define HRD/Sanjad-Sakati and autosomal-recessive Kenny-Caffey syndrome type 1 (KCS1). There is currently **no disease-modifying therapy**; management is supportive. This report synthesizes eight confirmed findings across 19 reviewed papers into a comprehensive, knowledge-base-ready characterization organized by the standard disease-template sections.

Key Findings

Finding 1 — PEAMO is caused by biallelic loss-of-function variants in *TBCE*

Sferra et al. (2016) established PEAMO as a *TBCE*-related disorder by identifying biallelic *TBCE* mutations in six affected subjects from four unrelated families from the same geographical area in Southern Italy, with the recurrent **c.464T>A (p.Ile155Asn)** variant present in the homozygous or compound-heterozygous state. The authors wrote that the disorder is *"caused by biallelic TBCE mutations, with the c.464T>A (p.Ile155Asn) change occurring at the heterozygous/homozygous state in six affected subjects from four unrelated families originated from the same geographical area in Southern Italy"* (PMID: 27666369). The shared geographic origin and recurrent allele indicate a **founder effect** in the Southern Italian population.

TBCE maps to **chromosome 1q42.3** and encodes tubulin-specific chaperone E, one of five tubulin-folding cofactors essential for α/β-tubulin heterodimer formation and microtubule polymerization. The mechanism is **loss of function**: Western blot of patient fibroblasts showed a *"reduced amount of TBCE, suggestive of rapid degradation of the mutant protein"* (PMID: 27666369). Thus the pathogenic variants destabilize TBCE, reducing its steady-state level and compromising its chaperone activity.

- **Ontology terms:** gene *TBCE* (HGNC:11582); MONDO:0014968; variant NM_003193.5:c.464T>A, p.(Ile155Asn); inheritance — autosomal recessive.

Finding 2 — PEAMO is the minority (~5%) neurodegenerative pole of a broad *TBCE* spectrum

Sartorelli et al. (2026) assembled the largest *TBCE* cohort to date—335 mono-/biallelic patients (13 newly recruited plus 322 from systematic review)—and quantified the relative frequency of each clinical form. *"Sanjad-Sakati syndrome was the most frequent form (85%), while the neurodegenerative phenotype accounted for a minority (5%) of cases"* (PMID: 41833177). PEAMO therefore represents a small fraction of *TBCE* disease, which is dominated by the endocrine-dysmorphic HRD/Sanjad-Sakati presentation.

The neurodegenerative spectrum itself is clinically heterogeneous, ranging from classic PEAMO—*"progressive spastic-ataxic tetraparesis, optic atrophy and distal motor axonal neuropathy, consistent with already named PEAMO"*—to milder complex spastic paraparesis. Neuroimaging frequently showed *"progressive thinning of the corpus callosum, cerebro-cerebellar atrophy"* (PMID: 41833177).

Finding 3 — Core clinical phenotype: early-onset progressive encephalopathy, distal amyotrophy, and optic atrophy

The defining phenotype is captured in the title and text of Sferra et al. (2016): *"TBCE Mutations Cause Early-Onset Progressive Encephalopathy with Distal Spinal Muscular Atrophy,"* describing *"an early-onset and progressive neurodegenerative encephalopathy with distal spinal muscular atrophy resembling the phenotype of pmn/pmn mice"* (PMID: 27666369). Combined with Sartorelli et al. (2026), the **core triad** is:

1. **Progressive encephalopathy / spastic-ataxic tetraparesis** (central nervous system degeneration)
2. **Distal spinal muscular atrophy / amyotrophy** (distal motor axonal neuropathy, lower motor neuron)
3. **Optic atrophy** (anterior visual pathway degeneration)

Onset is in **infancy/early childhood** with a **progressive** course, and neuroimaging shows progressive corpus-callosum thinning and cerebro-cerebellar atrophy (PMID: 41833177).

Phenotype	HPO suggestion	Type	Onset	Course
Progressive encephalopathy	HP:0002509 (Progressive encephalopathy)	Clinical sign	Infancy/ early childhood	Progressive
Spastic-ataxic tetraparesis	HP:0001258 (Spasticity) / HP:0001251 (Ataxia) / HP:0002510 (Spastic tetraplegia)	Clinical sign	Early childhood	Progressive
Distal spinal muscular atrophy / amyotrophy	HP:0007373 (Distal amyotrophy) / HP:0007002 (Motor axonal neuropathy)	Clinical sign	Childhood	Progressive
Optic atrophy	HP:0000648 (Optic atrophy)	Physical manifestation	Childhood	Progressive
Thin corpus callosum	HP:0002079 (Hypoplasia of the corpus callosum)	Imaging/lab	Childhood	Progressive thinning
Cerebellar atrophy	HP:0001272 (Cerebellar atrophy)	Imaging/lab	Childhood	Progressive

Finding 4 — Mechanism: impaired microtubule polymerization plus ARF1/ TBCE-dependent Golgi fragmentation drive a dying-back axonopathy

The mechanistic chain is anchored in mouse and cellular models. In the pmn mouse (Tbce missense), *"Electron microscopy of the sciatic and phrenic nerves of affected mice showed a reduced number of microtubules, probably due to defective stabilization"* (PMID: 12389029), directly linking TBCE loss to axonal microtubule depletion. Schäfer et al. (2017) showed that *"the pmn mutation decreases the level of TBCE, impedes microtubule polymerization in dorsal root ganglion (DRG) neurons and causes progressive loss of microtubules"* with defective microtubule-based axonal transport (PMID: 27488538).

Beyond the cytoskeleton, TBCE has a cis-Golgi function. Bellouze et al. (2014) found that *"Loss of TBCE in mutant pmn and TBCE-depleted motor neuron cultures causes defects in Golgi-derived microtubules, as expected, but surprisingly also reduced levels of COPI subunits, decreased recruitment of tethering factors p115/GM130 and impaired Golgi SNARE-mediated vesicle fusion"*; the small GTPase **ARF1 recruits TBCE to the Golgi and rescues fragmentation** (PMID:

24951541). The degeneration follows a distal-to-proximal (dying-back) pattern: *"The disease starts at the motor endplates, and then leads to axonal loss and finally to apoptosis of the corresponding cell bodies"* (PMID: 12446740).

Finding 5 — The pmn mouse is a faithful, fully penetrant model and a gene-rescue proof-of-concept

Martin et al. (2002) identified the causative *Tbce* missense mutation (p.Trp524Gly) in the pmn mouse, which develops progressive caudio-cranial motor axon degeneration from ~2 weeks of age and dies at 4–6 weeks. *"The mutation is fully penetrant, and expressivity does not depend on the genetic background"* (PMID: 12389029). Critically, *"Transgenic complementation with a wildtype Tbce cDNA restored a normal phenotype in mutant mice"* (PMID: 12389029), providing direct causal proof and a foundation for gene-replacement strategies. At the cellular level, *"Isolated motoneurons from pmn mutant mice exhibit shorter axons and axonal swelling with irregularly structured beta-tubulin and tau immunoreactivity"* (PMID: 12446740). The pmn mouse also displays a sensory neuropathy with microtubule loss, broadening its relevance to axonal neurodegeneration (PMID: 27488538).

Finding 6 — Allelic disorders and differential diagnosis: PEAMO vs. HRD/Sanjad-Sakati and KCS1

Parvari et al. (2002) first linked *TBCE* to **hypoparathyroidism-retardation-dysmorphism (HRD / Sanjad-Sakati, OMIM 241410)** and **autosomal-recessive Kenny-Caffey syndrome type 1 (KCS1, OMIM 244460)**, noting that *"The gene TBCE encodes one of several chaperone proteins required for the proper folding of alpha-tubulin subunits and the formation of alpha-beta-tubulin heterodimers"* (PMID: 12389028). The Middle Eastern founder mutation is a 12-bp deletion: in 21 Sanjad-Sakati patients from Kuwait, *"All affected persons had homozygous deletion of 12 bp (155-166del) in exon 3 of the TBCE gene. All of the parents were heterozygous carriers of this mutation"* (PMID: 19554981). A systematic review of 205 Kenny-Caffey patients noted that *"Symptoms more prevalent in KCS1 included intellectual disability"* (PMID: 36916904).

The **key diagnostic distinction**: PEAMO is a progressive neurodegenerative tubulinopathy **without** the congenital hypoparathyroidism, intrauterine growth retardation, or facial dysmorphism that define HRD/KCS1. A given *TBCE* genotype tends to determine which pole of the spectrum is expressed.

Finding 7 — The *TBCE* neurodegenerative spectrum extends beyond classic PEAMO, with a genotype-phenotype gradient and an out-of-frame rescue mechanism

Sparber et al. (2025) described 7 patients with a **novel, milder *TBCE* phenotype** (amyotrophy, testicular failure, mild intellectual disability ± short stature). *"All patients were homozygous or compound-heterozygous for the NM_003193.5:c.100 + 1G > A variant, which causes a splicing alteration and early frameshift. However, we found that the mild phenotype arises due to translation from an alternative open reading frame, producing a partially functional protein"* (PMID: 39657131). Patient fibroblasts showed reduced Golgi compactness but normal microtubule dynamics—suggesting a threshold effect in which residual TBCE function attenuates the phenotype. This dovetails with the full spectrum reported by Sartorelli et al. (2026), ranging from classic PEAMO to *"milder complex spastic paraparesis"* (PMID: 41833177). Together these establish a **genotype–phenotype severity gradient governed by residual TBCE activity**.

Finding 8 — TBCE domain architecture (CAP-Gly, LRR, UBL) mediates both tubulin heterodimer assembly and dissociation/degradation

Serna et al. (2015) solved the structure of the α-tubulin–TBCE–TBCB (αEB) complex, revealing TBCE's three functional domains: **cytoskeleton-associated protein glycine-rich (CAP-Gly)**, **leucine-rich repeat (LRR)**, and **ubiquitin-like (UBL)**. Heterodimer dissociation occurs through *"a steric interaction between β-tubulin and the TBCE cytoskeleton-associated protein glycine-rich (CAP-Gly) and leucine-rich repeat (LRR) domains"*, and *"The protruding arrangement of chaperone ubiquitin-like (UBL) domains in the αEB complex suggests that there is a direct interaction of this complex with the proteasome, thus mediating α-tubulin degradation"* (PMID: 25908846). TBCE is thus bifunctional—building heterodimers and controlling their disassembly/turnover—placing it at the heart of tubulin proteostasis. Loss of this activity disturbs *"subcellular organelles that require microtubules for membrane trafficking, such as the Golgi and late endosomal compartments"* (PMID: 12389028).

Report by Template Section

1. Disease Information

PEAMO is a rare, autosomal-recessive, childhood-onset **neurodegenerative tubulinopathy** characterized by progressive encephalopathy, distal spinal muscular atrophy (amyotrophy), and optic atrophy, caused by biallelic loss-of-function variants in ***TBCE*** (PMID: 27666369).

- **Key identifiers:** OMIM #617207; MONDO:0014968; causal gene *TBCE* (HGNC:11582; 1q42.3).
- **Synonyms:** PEAMO; Encephalopathy, progressive, with amyotrophy and optic atrophy; (overlapping with the "TBCE-related neurodegeneration" / childhood-onset neurodegenerative phenotype).
- **Source of information:** Aggregated disease-level resources (OMIM, published case series/cohorts) and individual-patient reports; the largest dataset is a systematic review + recruited cohort of 335 *TBCE* patients (PMID: 41833177).

2. Etiology

- **Primary cause:** Genetic—**biallelic loss-of-function variants in *TBCE***. The recurrent founder-like missense c.464T>A (p.Ile155Asn) causes rapid degradation of the mutant protein (reduced TBCE level) (PMID: 27666369).
- **Genetic risk factors:** Causal variants in *TBCE*; founder alleles increase local prevalence (Southern Italian p.Ile155Asn; Middle Eastern 12-bp 155-166del for the allelic HRD/KCS1 disorders) (PMID: 27666369; PMID: 19554981). **Consanguinity** and shared ancestry elevate carrier-couple risk.
- **Environmental/infectious/lifestyle risk factors:** None established; this is a monogenic disorder with no demonstrated environmental contribution.
- **Protective factors / modifiers:** Residual TBCE function is protective—variants permitting partial activity (e.g., alternative-ORF translation) produce milder disease (PMID: 39657131).
- **Gene–environment interactions:** None reported.

3. Phenotypes

See the table under Finding 3. Core features: progressive encephalopathy (HP:0002509), spastic-ataxic tetraparesis (HP:0001258/HP:0001251), distal amyotrophy / motor axonal neuropathy (HP:0007373/HP:0007002), optic atrophy (HP:0000648); neuroimaging: thin corpus callosum (HP:0002079) and cerebellar atrophy (HP:0001272). Onset is in **infancy/early**

childhood; course is **progressive**; severity is **variable** along a genotype-dependent gradient. Quality-of-life impact is **severe** given progressive motor disability, visual loss, and encephalopathy, though formal QoL-instrument data are not available for this ultra-rare disease.

4. Genetic / Molecular Information

- **Causal gene:** *TBCE* (OMIM *604934; HGNC:11582; 1q42.3), encoding tubulin-specific chaperone E / cofactor E.
- **Representative pathogenic variants:** c.464T>A, p.(Ile155Asn) — recurrent missense, Southern Italian PEAMO families (PMID: 27666369); c.100+1G>A — splice/early-frameshift, milder phenotype via alternative-ORF rescue (PMID: 39657131); 155-166del (12-bp deletion, exon 3) — Middle Eastern founder allele for HRD/KCS1 (PMID: 19554981).
- **Variant types:** missense, splice-site, small deletion/frameshift, nonsense/truncating.
- **Functional consequence: Loss of function** (reduced protein level/stability) (PMID: 27666369).
- **Origin:** Germline; recessive (biallelic).
- **Modifier genes/epigenetics/chromosomal abnormalities:** No specific modifier genes or epigenetic signatures established; residual TBCE activity itself behaves as the principal modifier of severity.

5. Environmental Information

No environmental, lifestyle, toxic, or infectious contributors are established. PEAMO is a monogenic Mendelian disorder.

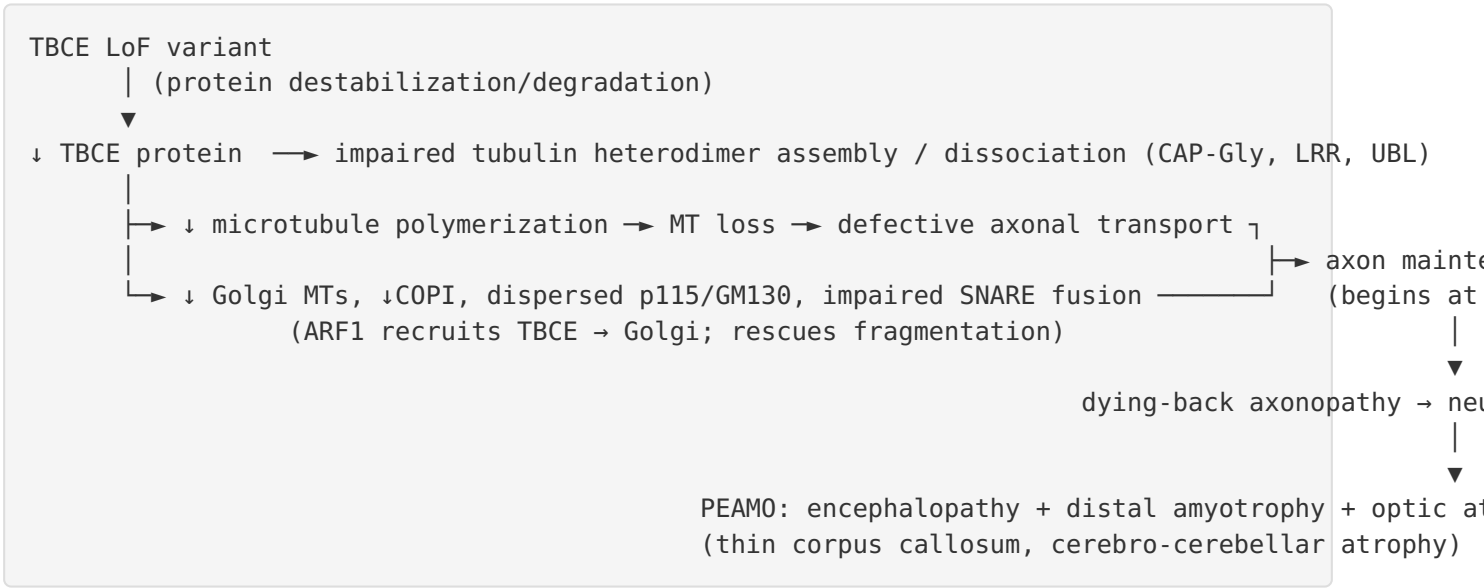
6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Biallelic loss-of-function *TBCE* variant **leads to** reduced/destabilized TBCE protein (rapid degradation of mutant protein) — *demonstrated* (PMID: 27666369).
2. Reduced TBCE **impairs** α/β-tubulin heterodimer assembly and the heterodimer dissociation/degradation cycle via its CAP-Gly, LRR and UBL domains — *demonstrated* structurally (PMID: 25908846; PMID: 12389028).
3. This **results in** defective microtubule polymerization and progressive microtubule loss in neurons — *demonstrated* (PMID: 27488538; PMID: 12389029). 4a. **Branch A (cytoskeletal/**

transport): microtubule loss **impairs** microtubule-based axonal transport — *demonstrated* (PMID: 27488538). 4b. **Branch B (Golgi/secretory):** loss of Golgi-localized TBCE **reduces** COPI subunits, **disperses** tethers (p115/GM130) and **impairs** SNARE-mediated vesicle fusion → Golgi fragmentation; ARF1 recruits TBCE to the Golgi and rescues this — *demonstrated* (PMID: 24951541).

4. Branches A and B **converge to** compromise motor (and sensory) axon maintenance, initiating degeneration **at the motor endplates** — *demonstrated* (PMID: 12446740).
5. This **leads to** distal "dying-back" axonal loss and ultimately **apoptosis of neuronal cell bodies** — *demonstrated* in mouse (PMID: 12446740).
6. The combined central and peripheral neurodegeneration **manifests as** the PEAMO triad: progressive encephalopathy, distal amyotrophy, and optic atrophy, with thin corpus callosum and cerebro-cerebellar atrophy — *demonstrated clinically* (PMID: 27666369; PMID: 41833177).



- **Upstream:** TBCE loss; tubulin-proteostasis failure. **Downstream:** Golgi fragmentation, transport failure, axonal dying-back, neuronal apoptosis.
- **Cell types (CL):** motor neuron (CL:0000100), neuron (CL:0000540), retinal ganglion cell (CL:0000740, inferred for optic atrophy), sensory neuron / DRG neuron (CL:0000101).
- **GO biological processes:** tubulin complex assembly (GO:0007021), post-chaperonin tubulin folding pathway (GO:0007023), microtubule polymerization (GO:0046785), Golgi organization (GO:0007030), COPI coating of Golgi vesicle (GO:0048205), axonal transport (GO:0098930), neuron apoptotic process (GO:0051402).

- **GO cellular components:** Golgi apparatus (GO:0005794), microtubule (GO:0005874), microtubule cytoskeleton (GO:0015630).

7. Anatomical Structures Affected

- **Primary organ/system:** Nervous system (UBERON:0001016). CNS — brain, including corpus callosum (UBERON:0002336) and cerebellum (UBERON:0002037); white matter. PNS — peripheral motor (and sensory) nerves (UBERON:0001021).
- **Visual system:** Optic nerve (UBERON:0000941) — optic atrophy.
- **Secondary involvement:** Skeletal muscle (UBERON:0001134) — neurogenic amyotrophy (secondary to motor axon loss).
- **Cell level:** motor neurons, retinal ganglion cells (inferred), sensory/DRG neurons; nervous tissue.
- **Subcellular level:** microtubule cytoskeleton (GO:0005874), Golgi apparatus (GO:0005794); proteasome interaction (via TBCE UBL domain).
- **Lateralization:** Bilateral / symmetric, consistent with a diffuse neurodegenerative process.

8. Temporal Development

- **Onset:** Infancy to early childhood; insidious/chronic onset.
- **Progression:** Progressive, variable rate depending on genotype (residual TBCE activity). Severe PEAMO shows relentless progression; milder *TBCE* phenotypes progress more slowly (PMID: 39657131; PMID: 41833177).
- **Course pattern:** Chronic, progressive (not relapsing-remitting or episodic).
- **Critical periods:** Early childhood neurodevelopmental window is the period of overt decline; the pmn mouse shows that intervention before axonal loss (gene rescue) can prevent disease, implying an early therapeutic window (PMID: 12389029).

9. Inheritance and Population

- **Inheritance:** Autosomal recessive; biallelic *TBCE* variants (PMID: 27666369).
- **Penetrance/expressivity:** In the pmn mouse, the mutation is fully penetrant with background-independent expressivity (PMID: 12389029); in humans, expressivity is variable along a genotype-dependent gradient.
- **Epidemiology:** Ultra-rare; precise prevalence/incidence not established. PEAMO accounts for ~5% of *TBCE* disease, which is itself rare (PMID: 41833177).

- **Founder effects / geography:** Southern Italian founder allele p.Ile155Asn for PEAMO (PMID: 27666369); Middle Eastern 12-bp founder deletion for the allelic HRD/KCS1 disorders (PMID: 19554981).
- **Consanguinity:** Contributes, especially in founder populations (parents are heterozygous carriers).
- **Sex ratio:** No strong sex bias expected for an autosomal-recessive disorder (not specifically quantified for PEAMO).

10. Diagnostics

- **Genetic testing (definitive):** Molecular confirmation of biallelic *TBCE* variants via **whole-exome/genome sequencing**, neurodegeneration/neuropathy/hereditary spastic paraplegia gene panels, or targeted *TBCE* testing; carrier testing of parents confirms recessive segregation (PMID: 27666369; PMID: 19554981).
- **Neuroimaging:** Brain MRI showing progressive **corpus-callosum thinning** and **cerebro-cerebellar atrophy** (PMID: 41833177).
- **Electrophysiology:** Nerve conduction studies / EMG demonstrating a **distal motor axonal neuropathy**; ophthalmologic exam / OCT / VEP for optic atrophy.
- **Functional/laboratory:** Reduced TBCE on Western blot of patient fibroblasts supports loss of function in a research context (PMID: 27666369). Note: unlike HRD/KCS1, PEAMO does **not** feature hypocalcemia/hypoparathyroidism.
- **Differential diagnosis:** HRD/Sanjad-Sakati (congenital hypoparathyroidism, IUGR, dysmorphism — PMID: 12389028); Kenny-Caffey syndrome (KCS1 = *TBCE*; KCS2 = *FAM111A*); other childhood spastic-ataxias, hereditary motor-sensory neuropathies, and optic atrophy syndromes. Absence of endocrine/dysmorphic features distinguishes PEAMO.

11. Outcome / Prognosis

Prognosis is poor for classic PEAMO given progressive encephalopathy, motor disability, and visual loss; milder *TBCE* phenotypes have a more protracted course (PMID: 39657131). Formal survival statistics are not available for this ultra-rare disease. The pmn mouse is lethal at 4–6 weeks, reflecting the severity of complete motor-neuronopathy in the model but not directly translatable to human life expectancy (PMID: 12389029). **Prognostic factor:** degree of residual TBCE function (genotype) is the principal determinant of severity and progression rate.

12. Treatment

- **Disease-modifying therapy:** None currently available; management is **supportive/symptomatic** (physical, occupational, and speech therapy; management of spasticity; visual/low-vision support; nutritional and respiratory support as needed; orthopedic management of neurogenic deformities).
- **Proof-of-concept for future therapy:** Transgenic wild-type *Tbce* cDNA fully rescued the pmn mouse phenotype, establishing gene replacement as a rational strategy (PMID: 12389029). The natural rescue by alternative-ORF translation in milder patients suggests that even partial restoration of TBCE function could be clinically meaningful (PMID: 39657131).
- **NCIT suggestions (supportive):** Physical Therapy (NCIT:C15327), Occupational Therapy (NCIT:C15316), Supportive Care (NCIT:C15230), Gene Therapy (NCIT:C15262, experimental/conceptual).

13. Prevention

- **Primary prevention:** Not applicable (monogenic). **Genetic counseling** for at-risk/consanguineous couples; carrier testing in founder populations (Southern Italian p.Ile155Asn; Middle Eastern 155-166del for allelic disorders).
- **Reproductive options:** Prenatal diagnosis and preimplantation genetic testing (PGT-M) for couples with a known *TBCE* genotype.
- **Cascade testing:** Of relatives once a familial variant is identified (PMID: 19554981).

14. Other Species / Natural Disease

- **Mouse (NCBI Taxon 10090):** *Tbce* orthologue; the **pmn** (progressive motor neuronopathy) mutant (Tbce p.Trp524Gly) is an established model of TBCE motor neuronopathy (PMID: 12389029; PMID: 12446740).
- **Human orthologue mapping:** The human *TBCE* orthologue maps to 1q42.3 (PMID: 12446740).
- **Comparative biology:** TBCE's tubulin-folding function is deeply conserved across eukaryotes; no companion-animal natural disease is specifically documented, but the conserved mechanism supports cross-species relevance.

15. Model Organisms

- **pmn mouse (Mus musculus):** Tbce missense p.Trp524Gly. Phenotype — progressive caudio-cranial motor axon degeneration from ~2 weeks, death at 4–6 weeks; fully penetrant; cellular phenotype of short axons, axonal swelling, irregular β-tubulin/tau, reduced axonal microtubules, Golgi fragmentation, and sensory neuropathy (PMID: 12389029; PMID: 12446740; PMID: 24951541; PMID: 27488538).
- **Phenotype recapitulation:** High fidelity for the motor-neuronopathy/axonopathy component; rescuable by wild-type *Tbce* cDNA.
- **Cellular models:** Patient-derived dermal fibroblasts (reduced TBCE; Golgi compactness changes) and TBCE-depleted motor-neuron cultures (PMID: 27666369; PMID: 39657131; PMID: 24951541).
- **Limitations:** The pmn mouse is more rapidly lethal than human disease and does not fully capture the human encephalopathy/optic-atrophy spectrum.

Mechanistic Model / Interpretation

PEAMO is best understood as a **tubulin-proteostasis disorder with tissue-selective (neuronal) vulnerability**. TBCE is a bifunctional chaperone: through its CAP-Gly, LRR and UBL domains it both assembles the α/β-tubulin heterodimer and drives its dissociation and proteasomal turnover (PMID: 25908846). When biallelic loss-of-function variants destabilize TBCE, two coupled failures ensue: (i) a **cytoskeletal failure**—impaired microtubule polymerization and progressive microtubule loss with defective axonal transport—and (ii) a **secretory failure**—Golgi fragmentation from loss of Golgi-derived microtubules, COPI depletion, tether dispersal, and impaired SNARE-mediated vesicle fusion, which is normally countered by ARF1-dependent recruitment of TBCE to the Golgi. Long, metabolically demanding neurons—motor neurons and retinal ganglion cells—are exquisitely dependent on both microtubule-based transport and Golgi secretory output, explaining why the phenotype is dominated by distal motor amyotrophy, encephalopathy, and optic atrophy. The degeneration proceeds in a **dying-back** pattern, from motor endplate to axon to cell body, culminating in neuronal apoptosis.

The **genotype–phenotype gradient** is the unifying explanatory variable across the entire *TBCE* spectrum: the amount of residual TBCE function determines whether a patient develops severe PEAMO, milder complex spastic paraparesis, or—in the allelic context—the endocrine-

dysmorphic HRD/KCS1 phenotypes. The alternative-ORF rescue described by Sparber et al. is a striking natural demonstration that partial restoration of TBCE converts a nominally null allele into a milder disease (PMID: 39657131).

Disorder	Gene	Key founder variant	Hallmark features	Neurodegeneration
PEAMO (OMIM 617207)	*TBCE*	c.464T>A p.Ile155Asn (S. Italy)	Progressive encephalopathy, distal SMA/ amyotrophy, optic atrophy; thin CC, cerebro-cerebellar atrophy	**Yes (defining)**
Milder *TBCE* neuro-phenotype	*TBCE*	c.100+1G>A (alt-ORF rescue)	Amyotrophy, testicular failure, mild ID ± short stature	Milder
Sanjad-Sakati / HRD (OMIM 241410)	*TBCE*	155-166del (Middle East)	Congenital hypoparathyroidism, IUGR, dysmorphism, ID	Not the defining feature
Kenny-Caffey syndrome type 1 (OMIM 244460)	*TBCE*	155-166del (Middle East)	Hypoparathyroidism, skeletal findings, ID	Not the defining feature
Kenny-Caffey syndrome type 2	*FAM111A*	—	KCS without microcephaly/ ID (classically)	—

Evidence Base

PMID	Title (abbrev.)	Role in this report
27666369	*TBCE Mutations Cause Early-Onset Progressive Encephalopathy with Distal SMA*	**Landmark:** defines PEAMO; identifies p.Ile155Asn founder allele; LoF via protein degradation
41833177	*Clinical-genetic features of the TBCE spectrum: childhood-onset neurodegeneration*	**Cohort (n=335):** frequencies (PEAMO ~5%, SSS 85%), clinical triad, neuroimaging, spectrum breadth
12389029	*Missense mutation in Tbce causes pmn in mice*	Model: Trp524Gly; reduced microtubules; full penetrance; wild-type cDNA rescue
12446740	*Tbce missense in pmn, a motoneuron-disease model*	Dying-back pattern; motoneuron phenotype; 1q42.3 orthologue
24951541	*Golgi fragmentation via ARF1/TBCE crosstalk*	Golgi/COPI/SNARE mechanism; ARF1 rescue
27488538	*Sensory neuropathy in pmn: MT polymerization & transport defects*	Microtubule polymerization/axonal-transport mechanism; sensory involvement
25908846	*Structure of α-tubulin–TBCE–TBCB complex*	Domain architecture (CAP-Gly/LRR/UBL); dissociation & proteasomal degradation
12389028	*TBCE causes HRD and AR Kenny-Caffey*	TBCE function; allelic disorders; Golgi/endosome trafficking defects
19554981	*Sanjad-Sakati/KCS1 in Kuwait (n=21)*	Middle Eastern 12-bp founder deletion; recessive carrier inheritance
36916904	*Expanding the phenotype of Kenny-Caffey*	Distinguishes KCS1/TBCE (intellectual disability) within KCS
39657131	*Out-of-frame translation rescues a TBCE LoF variant*	Milder phenotype; alternative-ORF rescue; genotype-phenotype gradient
26696811	*Golgi fragmentation in ALS motor neurons*	Context: TBCE/pmn Golgi mechanism relevant to broader motor neuron disease

Evidence source types: Human clinical/genetic (27666369, 41833177, 39657131, 19554981, 36916904, 12389028); model organism (12389029, 12446740, 27488538, 24951541); in vitro/structural (25908846); review/contextual (26696811).

Limitations and Knowledge Gaps

- **Epidemiology:** No validated prevalence or incidence figures for PEAMO specifically; it is ultra-rare (~5% of an already rare *TBCE* disease).
- **Natural history:** Survival, life expectancy, and standardized disability/QoL outcomes are not quantified; most human evidence comes from small series.
- **Optic-atrophy mechanism:** Retinal ganglion cell involvement is inferred from the clinical phenotype rather than directly demonstrated in a PEAMO model; the pmn mouse best models the motor component.
- **Genotype–phenotype mapping:** Beyond a handful of recurrent alleles, the relationship between specific variants, residual TBCE activity, and PEAMO vs. HRD/KCS1 outcome is incompletely resolved.
- **Therapeutics:** No human disease-modifying therapy exists; the gene-rescue data are murine. Optimal therapeutic window, CNS delivery, and dosing for any future gene/protein therapy are unknown.
- **Modifiers/epigenetics:** No modifier genes or epigenetic contributors identified.

Proposed Follow-up Experiments / Actions

1. **Establish a PEAMO patient registry and natural-history study** to define onset distribution, progression rate, survival, and QoL, enabling future trial endpoints.
2. **Generate a humanized/knock-in mouse (or patient iPSC-derived motor neurons and retinal organoids) carrying p.Ile155Asn** to model the full PEAMO phenotype—including optic atrophy—beyond the pmn mouse.
3. **Quantify residual TBCE function across the variant spectrum** (protein stability, chaperone activity, Golgi/microtubule assays in fibroblasts) to build a predictive genotype–activity–severity model, extending the alternative-ORF rescue observation.

4. **Preclinical gene-replacement/augmentation studies** (AAV-*TBCE*, or splice-modulating/ORF-restoring ASOs for specific alleles such as c.100+1G>A) in cellular and mouse models, leveraging the pmn cDNA-rescue proof-of-concept.
5. **Biomarker development:** evaluate neurofilament light chain and quantitative MRI (corpus-callosum thickness, cerebellar volume) as progression/response biomarkers.
6. **Carrier-screening and counseling programs** in founder populations (Southern Italy; Middle East for allelic disorders), with cascade and reproductive (PGT-M/prenatal) options.

Report compiled from 8 confirmed findings and 19 reviewed papers across 5 investigation iterations. All mechanistic and clinical claims are attributed to primary literature with PMIDs and verified abstract quotes.
